

SIGNALS AND SYSTEMS TEST 3

Number of Questions: 25

Time: 60 min.

Directions for questions 1 to 25: Select the correct alternative from the given choices.

- The pole zero pattern of a certain filter is shown in figure. The filter must be of the following type.
(A) Low pass (B) High pass
(C) Band pass (D) All pass
- Laplace transform of the function $f(t)$ is given by

$$-F(s) = \frac{4}{2s(2+s)}$$

The final value of the function $f(t)$ is

- (A) 0 (B) ∞
(C) 2 (D) 1
- Let $x[n] = \left(-\frac{1}{2}\right)^n u[n] - \left(-\frac{1}{3}\right)^n u[-n-1]$. The ROC (Region of Convergence) of the Z-transform of $x[n]$

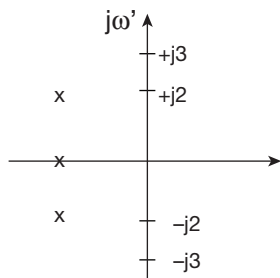
- $|Z| < \frac{1}{3}$
- $|Z| > \frac{1}{2}$
- $\frac{1}{3} < |Z| < \frac{1}{2}$
- None of these

- A sequence $x[n]$ with Z-transform $X(z) = Z^4 + Z^3 - 2Z + 2 - 3Z^{-5}$ is applied as an input to a linear, time-invariant system with impulse response $h[n] = 2\delta[n-4]$

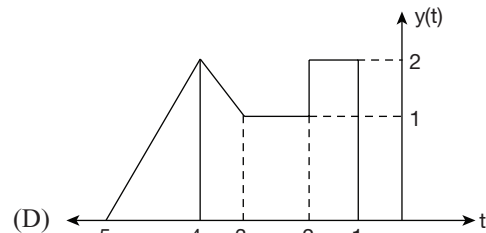
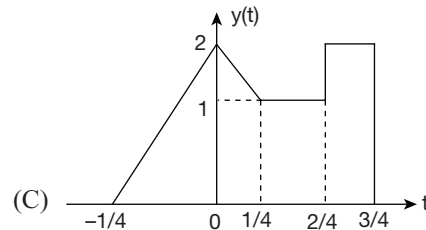
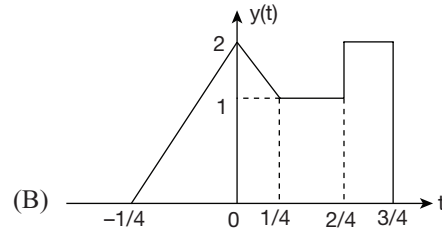
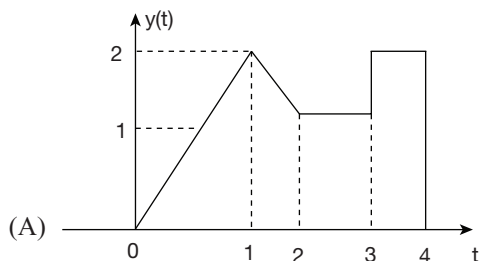
$$\text{Where } \delta[n] = \begin{cases} 1 & n=0 \\ 0 & \text{otherwise} \end{cases}$$

The output at $n = 3$ is

- 0
 - 2
 - +4
 - 4
- A continuous time signal is shown



The shifted version of the above signal $y(t) = x(t+4)$



- A system with input $x(t)$ and output $y(t)$ is described by the relation $y(t) = x(t+4) + x^3(t)$. This system is _____

- Linear and Time invariant
- Linear and Time-varying
- Non linear and time-invariant
- Non-linear and time-varying

- If, $x_1[n] = \cos\left(\frac{n\pi}{3}\right)$ and $x_2[n] = 5 \cos 3n +$

$$2 \sin\left(\frac{n\pi}{6}\right) \text{ then}$$

- $x_1[n]$ is a periodic, $x_2[n]$ is periodic
- Both $x_1[n]$ and $x_2[n]$ are periodic
- Both $x_1[n]$ and $x_2[n]$ are periodic
- $x_1[n]$ is periodic, $x_2[n]$ is a periodic

- The ROC of the Laplace transform of $x(t) = e^{-2t} u(t) + e^{-3t} u(t)$ will be

- $\text{Re}(s) > -2$
- $\text{Re}(s) > -3$
- $\text{Re}(s) < -3$
- $-2 < \text{Re}(s) < -3$

- If $X(s) = \frac{s+2}{s^2+4s+3}$, then $x(t)$ is

- $\frac{1}{2}e^{-t} + \frac{1}{2}e^{-3t}$
- $\frac{1}{2}e^{+t} - \frac{1}{2}e^{3t}$
- $\frac{1}{2}e^{-t} - \frac{1}{2}e^{-3t}$
- $E^{-t} - E^{-3t}$

10. Fourier transform of $e^{-2|t|}$ is

(A) $\frac{8}{16+\omega^2}$ (B) $\frac{8}{16-\omega^2}$
 (C) $\frac{4}{4+\omega^2}$ (D) $\frac{16}{64+\omega^2}$

11. Condition on the sampling interval T_s so that $x(t)$ is uniquely represented by the discrete time sequence $x[n] = x[nT_s]$

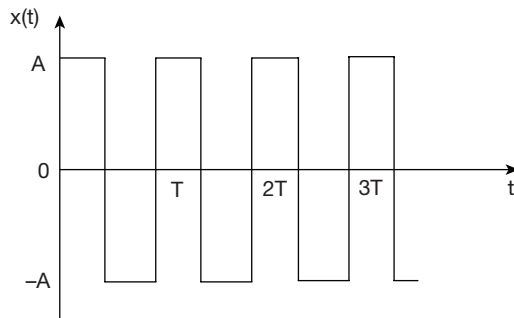
$x(t) = \cos(2\pi t) + 3\sin(6\pi t) + \sin(8\pi t)$

(A) $T_s < \frac{1}{8}$ (B) $T_s > \frac{1}{8}$
 (C) $T_s < \frac{1}{4}$ (D) $T_s > \frac{1}{4}$

12. A 2 kHz signal is sampled at the rate of 2.8 kHz and samples are applied to an ideal rectangular LPF with cut-off frequency 2.2 kHz then the output filter contains

- (A) only 400 Hz component
 (B) 400 Hz and 1000 Hz components
 (C) 400 Hz and 800 Hz components
 (D) 400 Hz, 800 Hz and 1000 Hz components

13. Laplace transform of periodic square wave



(A) $\frac{A}{S} \left(\frac{1-e^{-\frac{ST}{2}}}{1+e^{-\frac{ST}{2}}} \right)$ (B) $\frac{A}{S} \left(\frac{1-e^{\frac{+ST}{2}}}{1+e^{\frac{+ST}{2}}} \right)$
 (C) $\frac{A}{S} \left(\frac{1+e^{-\frac{ST}{2}}}{1-e^{-\frac{ST}{2}}} \right)$ (D) None of the above

14. The impulse response $h(t)$ of the system $y(t) = x(t - 2t_o)$ is, if all the initial condition are zero.

(A) $\delta(t - t_o)$ (B) $\delta(t - 2t_o)$
 (C) $\delta(t + t_o)$ (D) $\delta(t + 2t_o)$

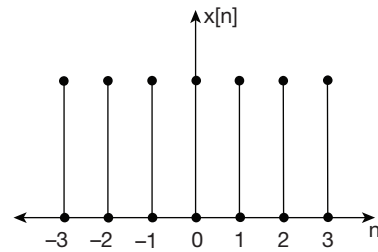
15. If an analog signal $x(t) = 5 \cos 75\pi t$. If the sampling frequency is 50 Hz, then the alias frequency corresponding to $f_s = 50$ Hz is

(A) 25 Hz (B) 50 Hz
 (C) 87.5 Hz (D) 100 Hz

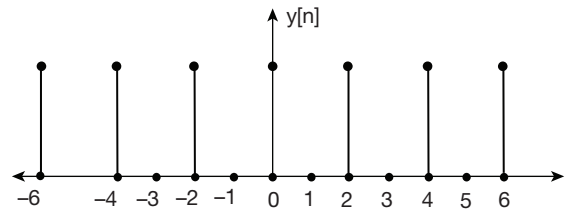
16. If $x_1(t) = 2 \cos 2\pi(30t)$ and $x_2(t) = 2 \cos 2\pi(60t)$ and 60 Hz signal is an alias of the 20 Hz signal, then the sampling frequency is

(A) 20 Hz (B) 30 Hz
 (C) 40 Hz (D) 60 Hz

17. If the DTFT of sequence $x[n]$ shown in figure is

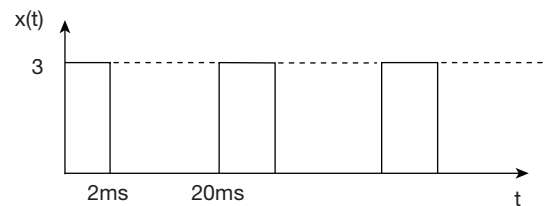


The DTFT $Y(e^{j\Omega})$ of sequence $y[n]$ shown will be _____



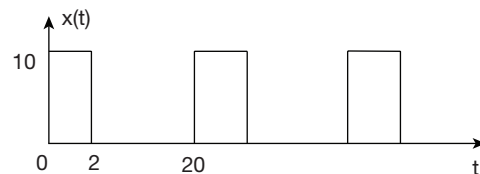
(A) $\frac{\sin 5\Omega}{\sin \Omega}$ (B) $\frac{\sin 1.25\Omega}{\sin 0.25\Omega}$
 (C) $\frac{\sin 10\Omega}{\sin 2\Omega}$ (D) None of the above

18. The constant component of the periodic pulse signal shown in figure is



(A) 0.6 (B) 0.3
 (C) 0.4 (D) None of these

19. Phase of the fundamental frequency component of the periodic pulse signal shown is



(A) $\angle -0.314$ rad (B) $\angle -0.314^\circ$
 (C) $\angle +0.314$ rad (D) $\angle +0.314^\circ$

20. Fourier series representation of $\cos^2 t$ is

(A) $-\frac{1}{4}e^{-j2t} + \frac{1}{2} - \frac{1}{4}e^{j2t}$ (B) $+\frac{1}{4}e^{-j2t} + \frac{1}{2} - \frac{1}{4}e^{j2t}$
 (C) $-\frac{1}{4}e^{-j2t} + \frac{1}{2} + \frac{1}{4}e^{j2t}$ (D) $\frac{1}{4}e^{-2jt} + \frac{1}{2} + \frac{1}{4}e^{2jt}$

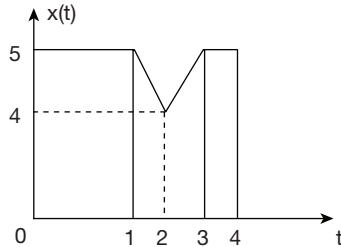
21. Differentiation of Signum function $s\left(\frac{d}{dt} 2\text{Sgn}(t)\right)$

- (A) $4 \delta(t)$ (B) $2 \delta(t)$
(C) $\delta(t)$ (D) $4 u(t)$

22. A periodic signal $x(t)$ is defined as $x(t) = (1 - t^4); 0 < t < T$. The Fourier coefficient b_n for the $x(t)$ is _____

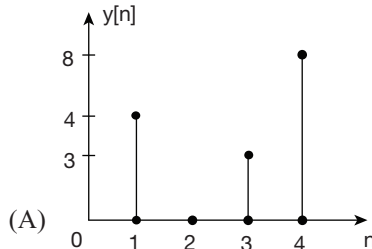
- (A) 2 (B) 1
(C) $\frac{1}{2}$ (D) 0

23. Energy of the signal shown in figure is

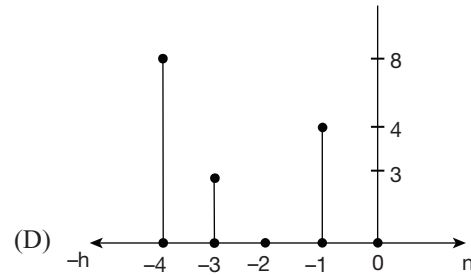
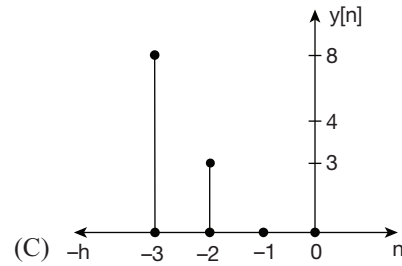
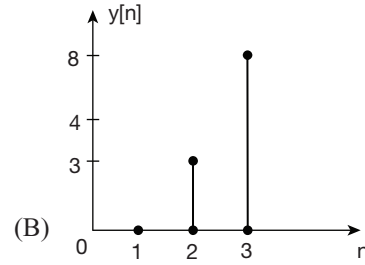


- (A) 91 Joules
(B) 95.5 Joules
(C) 19 Joules
(D) 31 Joules

24. The sequence $x[n]$ shown in figure whose DTFT is given as _____ $X(e^{j\Omega}) = 4 + 3e^{-2j\Omega} + 2e^{-3j\Omega}$
Which of the following signal corresponds to DTFT $Y(e^{j\Omega}) = 4 + 3e^{+2j\Omega} + 8e^{+3j\Omega}$



(A)



25. If $n\left(\frac{4}{5}\right)^{|n|} \stackrel{DTFT}{\leftrightarrow} X(e^{j\Omega})$

The Discrete time signal corresponds to $X(e^{j\Omega})$. $X(e^{j\Omega-\pi})$ is

- (A) $n^2 e^{j\pi n/2} \left(\frac{4}{5}\right)^{2|n|}$ (B) $n^2 \left(\frac{4}{5}\right)^{|n|} e^{j\pi(n/2)}$
(C) $n^2 \left(\frac{4}{5}\right)^{2|n|} e^{j\pi n}$ (D) $2\pi n^2 \left(\frac{4}{5}\right)^{2|n|} e^{j\pi n}$

ANSWER KEYS

- | | | | | | | | | | |
|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|
| 1. D | 2. D | 3. D | 4. D | 5. D | 6. C | 7. D | 8. A | 9. C | 10. D |
| 11. A | 12. A | 13. A | 14. B | 15. C | 16. B | 17. A | 18. B | 19. A | 20. D |
| 21. A | 22. D | 23. A | 24. C | 25. D | | | | | |

HINTS AND EXPLANATIONS

1. Since poles are parallel to $j\omega$ axis, So it is all pass filter.
Choice (D)

2. $\lim_{s \rightarrow 0} \lim_{s \rightarrow 0} SF(S)$

$$= \frac{4}{2s(2+s)} \cdot s = 1$$

Choice (D)

$$3. x[n] = \underbrace{\left(\frac{-1}{2}\right)^n}_{x_1[n]} u[n] - \underbrace{\left(\frac{-1}{3}\right)^n}_{x_2[n]} u[-n-1]$$

$$x_1[n] = \left(-\frac{1}{2}\right)^n u[n] \rightarrow \text{Right sided signal}$$

$$X_1(z) = \frac{1}{1 - \left(-\frac{1}{2}\right)Z^{-1}}$$

$$\text{ROC } |Z| \geq \frac{1}{2} \text{ or } |Z| > \frac{1}{2}$$

$$x_2[n] = \left(-\frac{1}{3}\right)^n u[-n-1] \rightarrow \text{Left sided signal}$$

$$X_2(z) = \frac{1}{1 - \left(-\frac{1}{3}\right)Z^{-1}} |Z| \leq \frac{1}{3}$$

Now, when both these transforms are added using linearity property –

$$\text{ROC will be } |Z| > \frac{1}{2} \text{ and } |Z| \leq \frac{1}{3}$$

So ROC does not exist

Choice (D)

$$4. Y(z) = H(z) X(z)$$

$$H(z) = 2 Z^{-4}$$

$$\therefore Y(z) = 2 Z^{-4} (Z^4 + Z^3 - 2Z^2 + 2 - 3Z^{-5})$$

$$= (2 + 2Z^{-1} - 4Z^{-3} + 4Z^{-4} - 6Z^{-9})$$

Taking Inverse Z-transform

$$y[n] = [2 + 2\delta(n-1) - 4\delta(n-3) + 4\delta(n-4) - 6\delta(n-9)]$$

$$\text{at } n=3, y[3] = -4$$

Choice (D)

$$5. x(t+4) \text{ is obtained by shifting } x(t) \text{ towards left by 4 time units.}$$

Choice (D)

$$6. y(t) = x(t+4) + x^3(t)$$

Linearity

$$x_1(t) \rightarrow y_1(t) = x_1(t+4) + x_1^3(t)$$

$$x_2(t) \rightarrow y_2(t) = x_2(t+4) + x_2^3(t)$$

$$ax_1(t) + bx_2(t) \rightarrow ax_1(t+4) + bx_2(t+4) + [ax_1(t) + bx_2(t)]^3$$

$$\neq ay_1(t) + by_2(t)$$

Hence the system is not linear _

Time Invariance

For a delayed input $x(t-t_o)$, output is $y(t, t_o) = x(t+4-t_o) + x^3(t-t_o)$

Now, the delayed output

$$y(t_1 - t_o) = x(t_1 - t_o + 4) + x^3(t_1 - t_o)$$

Hence this system is time-invariant.

Choice (C)

$$7. x_1[n] = \cos\left(\frac{n\pi}{3}\right) \cos\left(\frac{n\pi}{6}\right)$$

$$= \frac{1}{2} \cos\left[\frac{n\pi}{3} - \frac{n\pi}{6}\right] + \frac{1}{2} \cos\left[\frac{n\pi}{3} + \frac{n\pi}{6}\right]$$

$$= \frac{1}{2} \cos\left(\frac{n\pi}{6}\right) + \frac{1}{2} \cos\left(\frac{n\pi}{2}\right)$$

$$\text{Period of } \cos\left(\frac{n\pi}{6}\right):$$

$$N_1 = \frac{2\pi}{\Omega_1}(m) m \rightarrow \text{integer}$$

$$= \frac{2\pi}{\pi/6}(m) = 12 \text{ for } m=1$$

$$\text{Period of } \cos\left(\frac{n\pi}{2}\right):$$

$$N_2 = \frac{2\pi}{\Omega_2}(m) = \frac{2\pi}{\pi/2}(m) = 4 \text{ for } m=1$$

$$= 12 \text{ (for } m=3)$$

So period of $x_1[n] + x_2[n]$ is LCM (N_1, N_2), which is equal to $N=12$.

$$x_2[n] = 5\cos 3n + 2\sin\left(\frac{n\pi}{6}\right)$$

periodically of $\cos 3n \Omega_o = 3$

$$\frac{\Omega_o}{2\pi} = \frac{2}{2\pi} = \frac{3}{2\pi} \text{ (not rational)}$$

So $\cos 3n$ is not periodic, So $x_2[n]$ is also not periodic.

Choice (D)

$$8. x(t) = e^{-2t} u(t) + e^{-3t} u(t)$$

Since $u(t) = 1$, for $t > 0$; $x(t)$ is right sided signal.

The laplace transform of $x(t)$.

$$X(s) = \int_0^{\infty} e^{-2t} e^{-st} dt + \int_0^{\infty} e^{-3t} e^{-st} dt$$

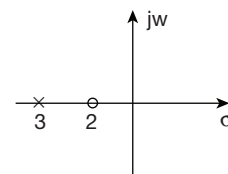
$$= \int_0^{\infty} e^{-(s+2)t} dt + \int_0^{\infty} e^{-(s+3)t} dt$$

Integral I converges if

$$R_e(s+2) > 0 \text{ or } R_e(s) > -2$$

Integral II converges if

$$R_e(s+3) > 0 \text{ or } R_e(s) > -3$$



Since $-2 > -3$ So $X(s)$ converges if $R_e(s) > -2$

Choice (A)

$$9. X(s) = \frac{s+2}{s^2+4s+3}$$

$$X(s) = \frac{(s+2)}{(s^2+4s+3)} = \frac{A}{s+1} + \frac{B}{s+3}$$

$$A = \frac{s+2}{s+3} \Big|_{s=-1} = \frac{1}{2}; \quad B = \frac{s+2}{s+1} \Big|_{s=-3} = -\frac{1}{2}$$

$$X(s) = \left[\frac{1}{2(s+1)} - \frac{1}{2(s+3)} \right]$$

$$x(t) = \left[\frac{1}{2} e^{-t} - \frac{1}{2} e^{-3t} \right] u(t)$$

Choice (C)

$$\begin{aligned}
 10. X(j\omega) &= \int_{-\infty}^{+\infty} e^{-8|t|} e^{-j\omega t} dt \\
 &= \int_{-\infty}^0 e^{8t} e^{-j\omega t} dt + \int_0^{\infty} e^{-8t} e^{-j\omega t} dt \\
 &= \int_{-\infty}^0 e^{(8-j\omega)t} dt + \int_0^{\infty} e^{(8+j\omega)t} dt \\
 &= \left[\frac{e^{(8-j\omega)t}}{8-j\omega} \right]_{-\infty}^0 + \left[\frac{e^{(8+j\omega)t}}{8+j\omega} \right]_0^{\infty} \\
 &= \frac{1}{8-j\omega} + \frac{1}{8+j\omega} = \frac{16}{64+\omega^2}
 \end{aligned}$$

Choice (D)

11. Maximum frequency ω_m present in $x(t)$ is _____

$$X(j\omega) = \begin{cases} 1 & |w| \leq 8\pi \\ 0 & |w| > 8\pi \end{cases}; \text{ And } W_m = 8\pi$$

$$\frac{2\pi}{T_s} > 16\pi; T_s < \frac{1}{8}$$

Choice (A)

12. A bandlimited signal should be sampled at least greater than the twice of the maximum signal. If this condition not satisfied aliasing will take place.
Sampling should be done at least at the rate of 4 kHz.
Sampling is done at 2.8 kHz
So -1.4 kHz to 1.4 kHz range does not cause aliasing Any frequency outside this range get aliased to this range.
 $f_m \pm kfs$
 $2 \pm 2.4 = \pm 0.4$ kHz
400 Hz signal will pass.
- Choice (A)

$$13. x(t) = \begin{cases} A & \text{for } t=0 \text{ to } \frac{T}{2} \\ -A & \text{for } t=\frac{T}{2} \text{ to } T \end{cases}$$

from periodicity property of Laplace transform _____
 $X(s) = L[x(t)]$
 If $x(t) = x(t + nT)$
 then $X(s) = \frac{1}{1-e^{-sT}} \int_0^T x_1(t) e^{-st} dt$, where $x_1(t)$ is one period of $x(t)$

$$= \frac{1}{1-e^{-sT}} \left[\frac{A}{s} \left(1 - e^{-\frac{sT}{2}} \right)^2 \right]$$

$$= \frac{1}{\left(1 + e^{-\frac{sT}{2}} \right) \left(1 - e^{-\frac{sT}{2}} \right)} \left[\frac{A}{s} \left(1 - e^{-\frac{sT}{2}} \right)^2 \right]$$

$$= \frac{A}{s} \left(\frac{1 - e^{-\frac{sT}{2}}}{1 + e^{-\frac{sT}{2}}} \right)$$

Choice (A)

14. By taking laplace transform with zero initial condition we get,

$$Y(S) = e^{-2S t_0} X(s)$$

$$H(s) = \frac{Y(s)}{X(s)} = e^{-2st_0}$$

Impulse response $h(t) = L^{-1} H(s)$

$$= L^{-1} \{ e^{-2st_0} \}$$

$$= \delta(t - 2t_0)$$

Choice (B)

$$15. x[n] = 5 \cos 75\pi t = 5 \cos 75\pi \times \frac{n}{F_s}$$

$$= 5 \cos \frac{75\pi \times n}{50} = 5 \cos \frac{3\pi n}{2}$$

Discrete time signal whose frequencies are separated by 2π are identical.

$$\therefore 5 \cos \frac{3\pi n}{2} = 5 \cos \left(\frac{3\pi}{2} + 2\pi \right) n = 5 \cos \frac{7\pi}{2} n$$

$$\text{Now } 5 \cos \frac{7\pi}{2} n \text{ is an alias of } 5 \cos \frac{3\pi}{2} n$$

So frequency of signal is $f = \frac{7}{4}$ cycles/sample

$$f = \frac{F}{F_s} \Rightarrow F = \frac{7}{4} \times 50 = 87.5 \text{ Hz}$$

So when $F_s = 50$ Hz, $F = 87.5$ Hz is an alias frequency.
Choice (C)

$$16. x_1(n) = 2 \cos \frac{2\pi(30 \times n)}{F_s}$$

Let the sampling frequency is $60 - 30 = 30$ Hz

$$x_1[n] = 2 \cos \left(\frac{30 \times n}{30} \right) 2\pi = 2 \cos 2n\pi$$

$$x_2[n] = 2 \cos 2\pi (60t)$$

$$= 2 \cos 2\pi \frac{60 \times n}{30} = 2 \cos 4n\pi = 2 \cos 2n\pi$$

$$\text{So } F_s = 30 \text{ Hz}$$

Choice (B)

17. $y[n]$ is time expanded version of signal $x[n]$

$$\text{so } y[n] = x \left[\frac{n}{2} \right]$$

Using time scaling property $x[n] \leftrightarrow X(e^{j\Omega})$

$$y[n] \leftrightarrow X \left[\frac{n}{2} \right] \leftrightarrow X(e^{j2\Omega}) = Y(j\Omega)$$

$$Y(e^{j\Omega}) = X(e^{j2\Omega}) = \frac{\sin 5\Omega}{\sin \Omega}$$

Choice (A)

18. The constant component of any periodic time domain signal is $\frac{a_0}{2}$

$$\text{So } a_0 = \frac{2}{T} \int_0^T x(t) dt$$

$$\text{So } x(t) = \begin{cases} 3, & \text{for } 0 \text{ to } 2\text{ms} \\ 0, & \text{for } 2 \text{ to } 20\text{ms} \end{cases}$$

$$\text{So } T = 20 \text{ ms}$$

$$\therefore a_o = \frac{2}{20} \int_0^2 3 dt$$

$$= \frac{1}{10} [3t]_0^2 = \frac{1}{10} [6 - 0]$$

$$\therefore \text{Constant component is } \frac{0.6}{2} = 0.3 \quad \text{Choice (B)}$$

19. The Fourier coefficient of n^{th} harmonic component is given by _____

$$C_n = \frac{1}{T} \int_0^T x(t) e^{-jn\Omega_o t} dt, \text{ where } \Omega_o = \frac{2\pi}{T}$$

The Fourier coefficient of fundamental component is obtained when $n = 1$

$$\therefore C_1 = \frac{1}{T} \int_0^T x(t) e^{-j\Omega_o t} dt = \frac{1}{20} \int_0^{20} 10 e^{-j\Omega_o t} dt$$

$$\therefore C_1 = \frac{1}{2} \left[\frac{e^{-\frac{j\pi t}{10}}}{-j\pi} \right]_0^{20}$$

$$= \frac{5}{-j\pi} \left[e^{-\frac{j\pi}{5}} - 1 \right] = \frac{5}{-j\pi} \left(\cos \frac{\pi}{5} - j \sin \frac{\pi}{5} - 1 \right)$$

$$= j1.5915 (0.8090 - j0.5878 - 1)$$

$$= 0.9355 - j0.304 = 0.9836 \angle -0.314 \text{ rad}$$

$$\text{Phase of fundamental component } C_1 = \angle -0.314 \text{ rad} \quad \text{Choice (A)}$$

$$20. \cos^2 t = \frac{\cos 2t + 1}{2}$$

$$= \frac{1}{2} + \frac{\cos 2t}{2} = \frac{1}{2} + \frac{1}{2} \frac{(e^{2t} + e^{-2t})}{2}$$

$$= \frac{1}{2} + \frac{1}{4} e^{-2jt} + \frac{1}{4} e^{-2jt}$$

$$= \frac{1}{4} e^{-2jt} + \frac{1}{2} + \frac{1}{4} e^{2jt}$$

Choice (D)

$$21. F[Sgn(t)] = \frac{2}{j\Omega}$$

$$F[2Sgn(t)] = \frac{4}{j\Omega}$$

$$\frac{d}{dt}[x(t)]^{FT} \leftrightarrow j\Omega \frac{4}{j\Omega}$$

$$\frac{d}{dt}[2Sgn(t)]^{F.T} \leftrightarrow 4$$

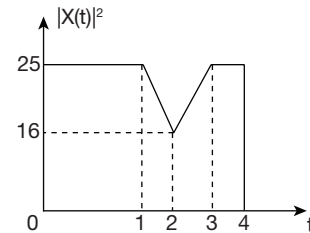
And we know that

$$\delta(t) \xleftrightarrow{F.T} 1$$

$$\text{So } \frac{d}{dt}[2Sgn(t)] = 4\delta(t) \quad \text{Choice (A)}$$

22. $x(t) = (1 - t^4); 0 < t < T$
now $x(-t) = [1 - (-t)^4]; 0 < t < T$
 $= 1 - t^4$, So the given signal even signal
For even signals $b_n = 0$ Choice (D)

$$23. \text{Energy } E = \int_{-\infty}^{+\infty} |x(t)|^2 dt$$



So $E = \text{Area of } x^2(t) = \text{Area of rectangle} - \text{Area of triangle}$

$$= 25 \times 4 - \frac{1}{2} \times (25 - 16) \times (3 - 1)$$

$$= 100 - \frac{1}{2} \times 9 \times 2$$

$$= 91 \text{ Joules}$$

Choice (A)

$$24. Y(e^{j\Omega}) = X(e^{j\Omega})$$

Using Time reversal property –

$$x[n] = X(e^{j\Omega})$$

$$x[-n] = X(e^{-j\Omega})$$

So $Y(e^{j\Omega})$ corresponds to DTFT of $x[-n]$

$$y[n] = x[-n]$$

Choice (C)

$$25. Y(e^{j\Omega}) = X(e^{j\Omega}) \times X(e^{j(\Omega-\pi)})$$

$$y[n] = 2\pi x[n] x_1[n]$$

$$x_1[n] = e^{j\pi n} x[n]$$

$$y[n] = 2\pi n^2 \left(\frac{4}{5} \right)^{2|n|} e^{j\pi n}$$

Choice (D)